

## VERTICAL LINKS

Sheathed (myelinated) nerve fibres are organized into bundles called projection tracts. These nerve fibres transmit impulses between the spinal cord and lower brain areas and the cerebral cortex above. The nerve tracts pass through a communication link called the internal capsule, and also

intersect the corpus callosum. In addition, similar bundles pass through the upper, outer zones of the white matter, from one area of the cerebral cortex to another. These bundles, called association tracts, convey nerve signals directly between different regions or centres of the cortex.

### Corona radiata

Zone where projection fibres spread out in a fan-like shape

### Grey matter (cerebral cortex)

Receives nerve impulses via projection fibres

### White matter

Contains both projection and association fibres

### Cranial nerves

### Internal capsule

A region of compact bands of nerve fibres

### PROJECTION PATTERN

The projection fibres pass through the upper part of the brainstem, then fan out and travel to the cerebral cortex.

## THE THALAMUS AND BRAINSTEM

The thalamus sits on top of the brainstem, and is shaped like two eggs placed side by side. It lies almost at the “heart” of the brain, and acts like a major relay station that monitors and processes incoming information before this is sent to the upper regions of the brain. The brainstem contains centres that regulate several functions vital for survival: these functions include the heartbeat, respiration, blood pressure, and some reflex actions, such as swallowing and vomiting.

### BRAINSTEM

The three main regions of the brainstem are the midbrain, pons, and medulla.

